Software Testing

EXP1: Identification of Software Defects in an ATM System

Aim: To examine the system specifications of an Automated Teller Machine (ATM) and identify, analyze, and document the software defects present in the system.

Features to be Tested

1. Card Validation
2. PIN authentication
3. Cash withdrawal
4. Cash dispensing

Bug List

Bug ID	Bug Name
ATM001	Invalid Card
ATM002	Invalid PIN
ATM003	Invalid Account Type
ATM004	Insufficient Balance
ATM005	Transaction Limit
ATM006	Daily Withdrawal Limit
ATM007	Invalid Denomination
ATM008	Receipt NOT Printed
ATM009	PIN Mismatch

BUG REPORT

Bug ID: ATM001
Bug Description: Invalid Card
Steps to Reproduce: Insert an invalid card into the ATM
Expected Result: Invalid card message should appear
Actual Result: Invalid card message appears
Status: Pass / Fail

Bug ID: ATM002
Bug Description: Invalid PIN
Steps to Reproduce: Insert a valid card and enter an incorrect PIN
Expected Result: Invalid PIN message should appear
Actual Result: Invalid PIN message appears
Status: Pass / Fail.

Bug ID: ATM003
Bug Description: Invalid account type
Steps to Reproduce: Log in and select an invalid account type.
Expected Result: Error message should appear.
Actual Result: Invalid account message appears.
Status: Pass / Fail.

Bug ID: ATM 004

Bug Description: Insufficient balance

Steps to Reproduce: Attempt to withdraw more than the available balance.

Expected Result: Transaction should be rejected.

Actual Result: Insufficient balance message appears.

Status: Pass/Fail.

Bug ID: ATM 005

Bug Description: Transaction limit exceeded

Steps to Reproduce: Enter an amount above the transaction limit

Expected Result: Transaction should be rejected.

Actual Result: Limit exceeded

Status: Pass/Fail.

Bug ID: ATM 009

Bug Description: PIN mismatch

Steps to Reproduce: Enter different PINs in the new PIN and confirmation field

Expected Result: PIN mismatch message should appear

Actual Result: PIN change is rejected

Status: Pass/Fail

Result: Nine potential software defects in the ATM system were identified and documented.

Exp No: 2

ARMS PORTAL - POSITIVE TESTING

Objectives

To prepare positive test case for the ARMS portal

Test Cases

Test Case ID : ARMS 001

Description : Valid Login

Steps : Enter valid username and password

Expected Result : Dashboard displayed

Actual Result : Dashboard displayed

Status : Pass / Fail

Test Case ID : ARMS 002

Description : View Profile

Steps : Open the profile section

Expected Result : Student details displayed

Actual Result : Students details displayed

Status : Pass / Fail

Test Case ID : ARMS 003

Description : View Attendance

Steps : Open the attendance section

Expected Result : Attendance displayed

Actual Result : Attendance displayed

Status : Pass / Fail

Test Case ID: ARMS004
Description: View Marks
Steps: Open the marks section
Expected Result: Marks displayed
Actual Result: Marks displayed
Status: Pass / Fail

Test Case ID: ARMS005
Description: View Results
Steps: Open the results sections
Expected Results: Results displayed
Actual Result: Results displayed
Status: Pass / Fail

Test Case ID: ARMS006
Description: Log out
Steps: Click Log out
Expected Result: Login Page displayed
Actual Result: Login Page displayed
Status: Pass / Fail

Result:
Six positive test cases were prepared for the ARMS portal.

Exp No: 3

ARMS PORTAL - NEGATIVE TESTING

AIM: To prepare negative test cases for the ARMS Portal

Test cases:

Test Case ID: ARMS001
Description: Invalid Username
Steps: Enter an incorrect username
Expected Result: Login rejected
Actual Result: Error message displayed
Status: Pass / Fail

Test Case ID: ARMS 002
Description: Invalid Password
Steps: Enter an incorrect password
Expected Result: Access denied
Actual Result: Login rejected
Status: Pass/ Fail

Test Case ID: ARMS 003
Description: Empty fields
Steps: Leave login fields empty and click Login
Expected Result: Validation msg displayed
Actual Result: Validation msg displayed
Status: Pass/Fail.

Test Case ID: ARMS 004

Description: Unauthorized access

Steps: Open the dashboard without login

Expected Result: Access denied

Actual Result: Login page displayed

Status: Pass / Fail

Test Case ID: ARMS005

Description: Marks Modification

Steps: Attempt to change marks using a student account

Expected Result: Modification rejected

Actual Result: Modification prevented

Status: Pass / Fail

Test Case ID: ARMS006

Description: Access After Logout

Steps: Log out and reopen dashboard

Expected Result: Login Required

Actual Result: Login page displayed

Status: Pass / Fail

Result: Six Negative Test cases were prepared for the ARMS Portal to verify input validation and security

Exp No: 4

E-Commerce Application - Positive Testing

Aim: To prepare positive test cases for an E-commerce Application

test cases:

Test Case ID: EC001

Description: User Registration

Steps: Enter valid details and submit

Expected Result: Account created successfully

Actual Result: Account created successfully

Status: Pass / Fail

Test Case ID: EC002

Description: User login

Steps: Enter valid login credentials

Expected Result: Login successful

Actual Result: Login successful

Status: Pass / Fail

Test Case ID: EC003

Description: Product Search

Steps: Search for an existing product

Expected Result: Product displayed

Actual Result: Products displayed

Status: Pass / Fail

Test Case ID: EC004

Description: Add to cart

Steps: Select a Product and click add to cart

Expected Result: Product added to cart

Actual Result: Product added to cart

Status: Pass | Fail

Test Case ID: EC005

Description: Successful Payment

Steps: Complete payment using valid test details

Expected Result: Payment successful

Actual Result: Payment successful

Status: Pass | Fail

Test Case ID: EC006

Description: Order Confirmation

Steps: Complete an order successfully

Expected Result: Order ID displayed

Actual Result: Order ID displayed

Status: Pass | Fail

Result: Six positive test case were prepared for E-Commerce Application

Exp No: 5

E-commerce Application-Negative Testing

Aim: To prepare negative test cases for an E-commerce Application

Test cases

Test case ID: EC001

Description: Invalid Email

Steps: Register using an invalid Email

Expected Result: Registration rejected

Actual Result: Error message displayed

Status: Pass / Fail

Test case ID: EC002

Description: Empty Fields

Steps: Submit registration with empty fields

Expected Result: Validation msg displayed

Actual Result: Validation msg displayed

status: Pass / Fail

Test Case ID: EC003

Description: Invalid Login

Steps: Enter an incorrect password

Expected Result: Login rejected

Actual Result: Error msg displayed

Status: Pass / Fail

Test case ID: EC004
Description: Out of stock Product
Steps: Attempt to purchase an unavailable product
Expected Result: Purchase rejected
Actual Result: Product unavailable msg
Status: Pass / Fail

test case ID: EC005
Description: Failed payment
steps: Make a payment using a failing test transaction
Expected Result: Payment Rejected
Actual Result: Payment failure msg displayed
Status: Pass / Fail

Test Case ID: EC006
Description: Empty cart Checkout
Steps: Click Checkout with an empty cart
Expected Result: Checkout prevented
Actual Result: Checkout prevented
Status: Pass / Fail

Result: Six Negative test cases were prepared for the E-Commerce Application to verify input validation and error handling.